HDRSL Net for Accurate High Dynamic Range Imaging-Based Structured Light 3D Reconstruction

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***Abstract*—In fringe projection profilometry systems, accurately reconstructing 3D objects with varying surface reflectivity requires high dynamic range (HDR) imaging. However, the limited dynamic range of single-exposure cameras poses challenges for capturing HDR fringe patterns efficiently. This paper introduces a deep learning-based HDR structured light 3D reconstruction pipeline, comprising an HDR Fringe Generation Module and a Phase Calculation Module. The HDR Fringe Generation Module employs an end-to-end network with attention guidance and feature distillation to reconstruct HDR fringe images from short- and long-exposure low dynamic range (LDR) inputs. The Phase Calculation Module processes the phase information from HDR fringes to enable 3D reconstruction. On a metallic HDR dataset, the method achieved a phase error of 0.105, comparable to the 4-exposure 6-step Phase Shifting Profilometry (PSP) method (0.069), with only 8.3% of the projection time. Experimental results demonstrate the robustness of our approach under diverse object geometries, exposure levels, and challenging global illumination environments. In quantitative measurements, our method achieved accuracies of sub-50μm on ceramic spheres, flat plates and metal step object. Ablation experiments confirmed that feature distillation and attention module effectively enhance the HDR Fringe Generation Module, producing high-quality HDR fringe patterns critical for reconstructing objects with HDR surface reflectivity. Furthermore, we constructed an HDR imaging metal dataset comprising 1,700 samples of machined metal parts with diverse shapes, sizes, and materials, making it a benchmark in the field of HDR structured light measurement. Our method offers a general HDR imaging-based structured light 3D reconstruction approach, integrating the two modules into an efficient, end-to-end solution for objects with HDR reflective surfaces.**

***Index Terms*—Structured light 3D measurement, high dynamic range imaging, phase calculation network, feature distillation, attention guidance.**

Received 30 December 2024; revised 30 May 2025; accepted 13 August 2025. Date of publication 25 August 2025; date of current version 28 August 2025. This work was supported in part by Shenzhen Science and Technology Program under Grant JCYJ20240813112003005 and in part by the Major Key Project of Peng Cheng Laboratory under Grant PCL2023A09. The associate editor coordinating the review of this article and approving it for publication was Prof. Yipeng Liu. *(Corresponding authors: Xiaojun Liang; Xinghui Li.)*

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Data is available on-line at https://github.com/Wangh257/HDRSL

Digital Object Identifier 10.1109/TIP.2025.3599934

I. Introduction

3D MEASUREMENT methods are widely applied in the measurement and inspection of industrial parts [1], [2]. Among them, Fringe Projection Profilometry (FPP) [3], as an active measurement method, is characterized by high speed [4], high accuracy [5], high resolution [6], [7], and strong anti-interference capabilities [8], making it widely used. A typical FPP system consists of a projector and a camera, forming a binocular measurement system [9]. After system calibration [10], [11], the 3D coordinates of an object can be determined based on the relationship between the camera and the projector. The projector projects fringe patterns encoded with phase information, and the camera captures the modulated fringe patterns reflected by the object's surface morphology [12], [13]. Therefore, accurately capturing modulated fringe patterns is essential for accurate measurements [14]. However, the limited dynamic range of a single exposure makes it difficult to image objects with varying surface reflectivity: high-reflectivity surfaces may cause saturation, while low exposure leads to poor signal-to-noise ratio (SNR) in dark regions [15], [16], [17].

To address this issue, various solutions have been proposed [18], [19]. From a software perspective, without changing the system parameters, post-processing algorithms can correct fringe patterns and mitigate errors caused by image saturation [19], [20], [21]. In FPP systems, sinusoidal fringes are typically used for measurement, requiring at least three accurate sampling points per fringe cycle. Typical approaches to increase sampling include adding more projection steps [19] or using inverted patterns [21], which help balance saturation-induced errors. However, these methods project more fringe patterns, reducing measurement efficiency, introducing motion errors, and failing to eliminate imaging errors, resulting in limited accuracy improvements.

From the hardware perspective, some devices can directly capture high dynamic range (HDR) images [22], but they are typically expensive. Alternatively, polarization-based methods suppress reflections using polarimetric cameras [18], but they require extra hardware and often suffer from low resolution and poor image quality. Multi-exposure techniques [23], [24], [25], are widely adopted. By combining long and short exposure images, they generate HDR fringes that cover both low- and high-reflectivity regions. A similar effect can be achieved by adjusting the intensity of the projection. To ensure high-quality HDR images, multiple exposure levels are required,

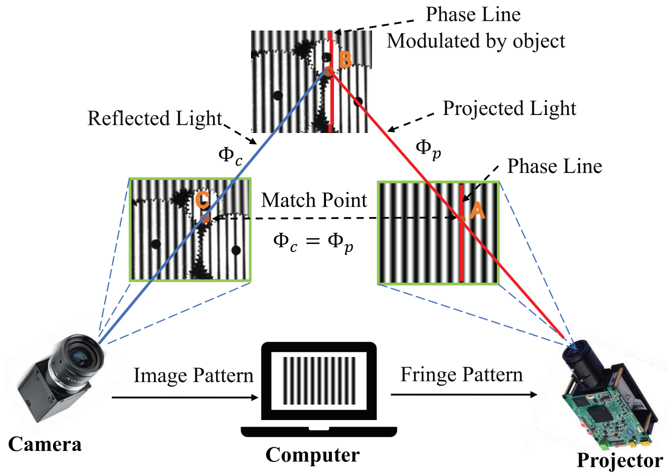

Fig. 1. Structure of FPP measurement. The projector projects fringe patterns containing phase information, which are modulated by the surface of the measured object. The camera captures and demodulates the patterns to reconstruct the 3D surface profile of the object.

which reduces efficiency and necessitates a trade-off between accuracy and speed.

In recent years, deep learning techniques have generated richly colored HDR images from a few low dynamic range (LDR) images [26], [27]. However, the HDR images generated by these methods lack the phase information, making it unsuitable for 3D reconstruction tasks. Zhang et al. [16] was the first to apply deep learning to HDR reconstruction, using the results of 12-step PSP method as ground truth (GT) to supervise a three-step process, simulating an increase in the number of steps. However, 12-step also faces challenges with HDR imaging, limiting its accuracy. HDRNet [28], Y-FFC [29] and DC-UNET [30] trained neural networks using simulated data to separate specular reflections from diffuse reflections in high-reflectivity images, enabling reconstruction of reflective surfaces. This approach faces practical limitations due to difficulties in accurate reflection separation and generalization from synthetic data. Liu et al. introduced SP-CAN [31], which reconstructs HDR images from a single short-exposure fringe image, simulating the multi-exposure approach with optimized exposure time. However, the single LDR fringe image lacks sufficient features, and the optimal exposure time must still be calculated. The above methods require correction for each LDR fringe image, followed by phase analysis with traditional PSP methods, making the process inefficient. Currently, deep learning has been effectively applied to phase analysis. For example, Feng et al. [32] used CNNs to separate the background and fringe patterns from single-frame images. Song et al. proposed SRPRNet [33], which quickly achieves super-resolution phase retrieval using a single pattern. In addition, deep learning methods are highly data-dependent. On the data front, Li et al. [10] and Qian et al. [34] introduced datasets containing approximately 1,000 gypsum samples. However, the structured light field still lacks large-scale HDR imaging datasets.

In this paper, we introduce a deep learning pipeline into the HDR imaging-based structured light measurement system. This pipeline generates HDR fringe patterns from LDR patterns captured under short and long exposure. Furthermore, by leveraging a neural network for phase analysis, the conventional process involving multiple steps is reduced to a single-step strategy. This method retains the high accuracy of the traditional multi-exposure, multi-frequency and multi-step PSP method while significantly accelerating 3D reconstruction. On the metal dataset, our method achieved a phase error of 0.105, which is close to the traditional PSP method, while using only 8.3% of the projection time. It also achieved an accuracy of sub-50 m on standard spheres, plate, and metal steps. Experimental results demonstrate that our approach is a general HDR imaging-based structured light 3D reconstruction method for objects with HDR reflective surfaces.

In addition, we onstructed the HDR imaging metal dataset developed in this study. This dataset contains 1,700 samples of machined metal parts with diverse shapes, sizes, and materials, making it a benchmark in the field of HDR structured light measurement.

The second section will introduce the principles of Multi-exposure FPP. The third section presents the experiments and analysis, and the final section concludes the paper.

II. Principle

A. Principles of Fringe Projection Profilometry

FPP is a binocular vision technique that models the projector and camera using optical centers and pixel rays [12]. As shown in Fig. 1, the red and blue rays (AB and BC) represent the viewing directions of the projector and camera, respectively, and the depth is determined by their intersection. With calibration of both the camera [35] and projector [36], the 3D coordinates $(X_c; Y_c; Z_c)$ in the camera coordinate system can be formulated as a function of the projector's horizontal pixel coordinate x_p [10], [12], [37], as shown in Eq. 1.

$$X_c = \frac{1}{f_x(x_p)} + c_x; \quad Y_c = \frac{1}{f_y(x_p)} + c_y;$$
$$Z_c = \frac{1}{f_z(x_p)} + c_z \tag{1}$$

Here, f_x, f_y, f_z, c_x, c_y, and c_z are system calibration parameters. To determine pixel correspondence, the projector emits phase-shifted sinusoidal fringes, and the camera captures their deformation on the object surface.The intensities of the projected and captured fringes are given by:

$$I_p(x_p; y_p) = B_p \cos \ \Phi_p(x_p; y_p) \tag{2}$$
$$I_c(x_c; y_c) = A_c + artI_p(x_p; y_p) + \Delta I_n \tag{3}$$

where $(x; y)$ represents the pixel coordinates, A is the constant ambient light intensity, B is the amplitude of the projected light, and Φ is the phase. ΔI_n represents random noise, a represents the camera response parameter, r is the reflectivity of object, and t is the exposure time. To retrieve the phase, N-step phase-shifted fringes are projected, and the captured images are given by:

$$I_c^k = A_c + B_c \cos \ \Phi_c + \frac{2 \ k}{N}$$
$$(k = 0; 1; :::; N \ \ 1) + \Delta I_n \tag{4}$$

Assuming A_c remains constant across N steps, $I_c^k(x, y)$ forms an N-point discrete signal. The fringe amplitude and initial phase are then obtained via Fourier transform:

$$F_{(f=1)}(I_c) = C + j \cdot S \tag{5}$$

$$S = \frac{2}{N}\sum_{k=0}^{N-1} I_c^k \cdot \sin\left(\frac{2\pi k}{N}\right); \quad C = \frac{2}{N}\sum_{k=0}^{N-1} I_c^k \cdot \cos\left(\frac{2\pi k}{N}\right) \tag{6}$$

$$\varphi_c = \tan^{-1}\left(-\frac{S}{C}\right); \quad B_c = \sqrt{S^2 + C^2} \tag{7}$$

Here, S and C represent the sine and cosine components of the fringe frequency, respectively, and F denotes the Fourier transform. Due to the periodicity of $\tan^{-1}$, the computed phase φ_c is wrapped within $[-\pi, \pi]$ and requires unwrapping [38], [39]. Phase unwrapping relies on lower-frequency wrapped phases to anchor higher-frequency ones. The relationship between the low- and high-frequency phases is given by:

$$\Phi_{\text{high}} = \varphi_{\text{high}} + 2\pi \cdot \text{round}\left(\frac{n\Phi_{\text{low}} - \varphi_{\text{high}}}{2\pi n}\right) \tag{8}$$

where n denotes the frequency ratio between the high and low frequencies. Low-frequency fringes offer strong noise robustness, while high-frequency fringes ensure high accuracy. Phase unwrapping is performed hierarchically, and the highest-frequency phase is used for 3D reconstruction. The recovered phase Φ has a linear relationship with the projector coordinate x_p. Combined with Eq. 1, the spatial coordinate exhibits a one-to-one correspondence with the phase, which can be modeled using a third-order polynomial [10], [12], [29], [37].

$$\begin{aligned} X_c &= a_1\Phi_c^3 + b_1\Phi_c^2 + c_1\Phi_c + d_1, \\ Y_c &= a_2\Phi_c^3 + b_2\Phi_c^2 + c_2\Phi_c + d_2, \\ Z_c &= a_3\Phi_c^3 + b_3\Phi_c^2 + c_3\Phi_c + d_3. \end{aligned} \tag{9}$$

During calibration [37], the absolute phase and 3D coordinates of the calibration board at over 12 positions are computed pixel-wise. The polynomial coefficients a, b, c, and d are then fitted via least squares and stored in a $(H, W, 12)$ lookup table to enable fast 3D reconstruction.

B. Principle of Multi-Exposure

As accurate phase retrieval is key to 3D reconstruction, the phase error caused by intensity noise ΔI_n can be derived based on Eq. 4, Eq. 6, and Eq. 7, and is expressed as:

$$\Delta\varphi_c = \frac{2}{NB_c}\sum_{n=1}^{N} \sin\left(\frac{2\pi k}{N} - \varphi\right) \cdot \Delta I_n \tag{10}$$

The equation shows that when ambient and projected light intensities remain constant, the error is inversely related to the imaged fringe amplitude. Low amplitude, typically due to low surface reflectivity or short exposure times, results in increased phase error [15] as shown in Fig. 2(a). Increasing the fringe amplitude improves accuracy, but excessive amplitude may cause pixel saturation, as shown in Fig. 2(b), resulting in information loss and reconstruction failure. when surface reflectivity varies significantly, balancing saturation and low SNR with a single exposure is challenging.

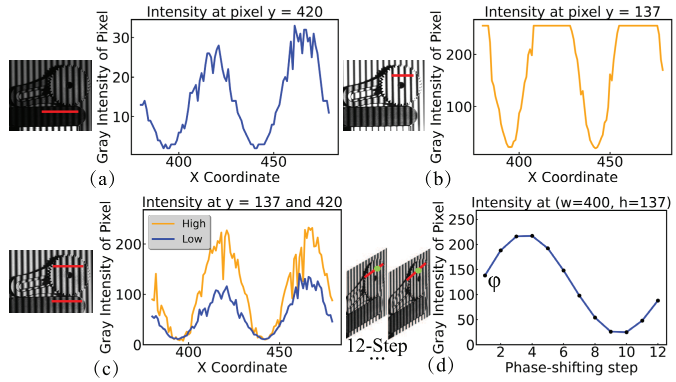

Fig. 2. (a) Fringe pattern in low reflectivity regions with short exposure time, (b) Fringe pattern in high reflectivity regions with long exposure time, (c) HDR fringe pattern generated from multiple exposures, (d) HDR fringe intensity variation in single-pixel 12-step phase shifting.

In multi-exposure structured light, a series of fringe patterns captured under different exposure levels $\{I_{m,n}^t(x, y)\}$ are used for 3D reconstruction, where $t = 0, 1, \ldots, T-1$ denotes the exposure index, $m = 0, 1, \ldots, M-1$ is the frequency index, and $n = 0, 1, \ldots, N-1$ is the phase-shift index. For each fringe image, the highest possible non-saturated exposure level t^{nosat} is selected. The final HDR fringe is then defined as:

$$I_{m,n}^{\text{HDR}}(x, y) = I_m^{\min\left(t_0^{\text{nosat}}, t_1^{\text{nosat}}, \ldots, t_{N-1}^{\text{nosat}}\right)}(x, y) \tag{11}$$

The multi-exposure strategy suppresses saturation and enhances SNR, as shown in Fig. 2(c). Selecting the minimum non-saturated exposure across the N phase steps ensures fringe sinusoidality and accurate wrapped phase recovery, as shown in Fig. 2(d). Clearly, there is a trade-off between acquiring high-quality HDR fringes and maintaining measurement efficiency [40]. To address this, we propose a neural network-based approach that simplifies both HDR fringe generation and phase retrieval.

III. Proposed Method

The conventional multi-exposure PSP method first synthesizes HDR fringe images and then computes phase maps via phase-shifting, achieving high accuracy but with low efficiency. Motivated by this pipeline, we propose the *HDR Structured Light 3D Reconstruction Network* (HDRSL-Net), which leverages the image-to-image modeling capability of CNNs, as illustrated in Fig. 3(a). The network consists of two modules: an HDR Fringe Generation Module that reduces the number of exposures from T to 2, and a Fringe Phase Calculation Module that reduces the number of phase-shifting steps from N to 1. In addition, M frequencies are still required for reliable phase unwrapping. As a result, our end-to-end method maintains the high accuracy of traditional multi-exposure PSP while reducing the total number of fringe images from $T \times M \times N$ to $2 \times M \times 1$.

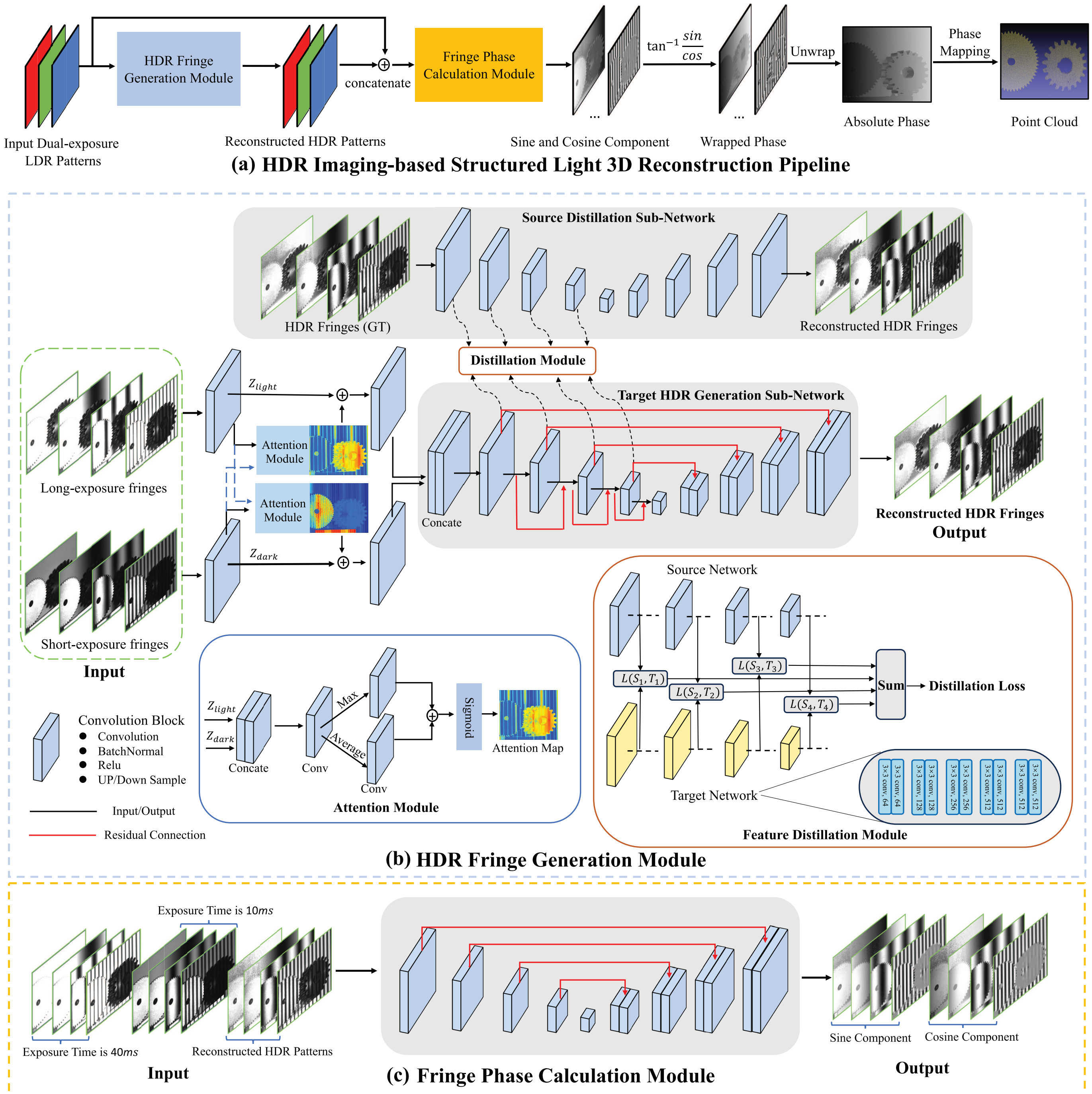

Fig. 3. The structure of the pipeline for 3D structured light measurement under HDR imaging conditions. The HDR Fringe Generation Module, an end-to-end sub-network enhanced by attention guidance and feature distillation, reconstructs HDR fringe patterns from short- and long-exposure LDR inputs. The Phase Calculation Module subsequently computes the sine and cosine components of the fringe phase.

A. HDR Fringe Generation Module

HDR fringe generation is formulated as a pixel-wise regression task. We adopt an end-to-end U-Net architecture with residual connections as the backbone of the HDR generation module, as illustrated in Fig. 3(b). It consists of five downsampling encoding blocks and five upsampling decoding blocks. The input consists of the first-step fringe images under the shortest and longest exposures across M frequencies, providing complementary features for high- and low-reflectivity regions. The output is the corresponding HDR fringe for each frequency. To guide the network in learning the exposure characteristics of the structured light system, we use the multi-exposure fused fringe (Eq. 11) as the ground truth. To achieve this, we impose output-space constraints by employing a combination of loss functions, including mean squared error (MSE), mean absolute error (MAE), and structural similarity (SSIM) [41].

$$L_1 = \frac{1}{M \quad H \quad W} \sum_{m=1} \sum_{i=1} \sum_{j=1} \tilde{I}_{mij} \quad I_{mij} \tag{12}$$

$$L_2 = \frac{1}{M \cdot H \cdot W} \sum_{m=1}^{M} \sum_{i=1}^{H} \sum_{j=1}^{W} \left(\tilde{I}_{mij} - I_{mij} \right)^2 \tag{13}$$

$$L_{\text{output}} = L_1 + \sqrt{L_2} + \left(1 - \text{SSIM}(\tilde{I}_m; I_m)\right) \tag{14}$$

where M is the number of fringe frequencies, W and H are the image width and height, $\tilde{I}$ is the network output, and I is the ground truth.

1) Feature Distillation: Since the generated HDR fringe maps are used for downstream phase retrieval, we introduce a feature distillation mechanism in addition to the output-space constraint loss L_{output}. As shown in Fig. 3(b), the backbone is treated as a student model, and a teacher model with a similar architecture and identical loss function is constructed. The teacher takes ground-truth HDR fringes as input and output, providing intermediate features to guide the student toward preserving phase-relevant information. To ensure spatial consistency, we apply an L_2 loss to constrain feature distances and a cosine loss to align feature directions. Distillation is applied to four encoder layers, and the final loss function L_{all} combines the output-space loss and the feature-space distillation loss:

$$L_2 = \frac{1}{L \cdot L_i} \sum_{l=1}^{L} \sum_{i=1}^{L_i} \left(a_s(i) - a_t(i)\right)^2 \tag{15}$$

$$L_{angle} = \frac{1}{L} \sum_{l=1}^{L} \left(1 - \frac{\text{vec}(a_s)^T \cdot \text{vec}(a_t)}{\|\text{vec}(a_s)\| \|\text{vec}(a_t)\|}\right) \tag{16}$$

$$L_{all} = L_2 + L_{angle} + L_{output} \tag{17}$$

Here, L is the number of distillation layers (set to 4), and L_i denotes the number of pixels at the i-th layer. a is the vectorized feature map, with subscripts s and t indicating the student and teacher networks, respectively.

2) Attention Module: To enhance the network's ability to model HDR regions, we introduce an attention module into the HDR generation module. It guides shallow features to focus on low-exposure cues in high-reflectivity regions and high-exposure cues in dark areas. The attention structure is illustrated in Fig. 3(b). The module concatenates short- and long-exposure features (Z_{light}, Z_{dark}), and processes them through three 3×3 convolutional layers with 64 filters and ReLU activations. The last two layers extract mean and max features, which are fused and passed through a sigmoid function to generate attention weights in the range $[0, 1]$. The trained attention weights guide and enhance task-relevant features in the LDR images.

B. Fringe Phase Calculation Module

To enable efficient phase retrieval from HDR fringe patterns, we designed a Phase Calculation Module that directly analyzes the HDR fringe images. For simplicity, we employed a U-Net architecture, which has proven effective in phase analysis tasks [32], [39]. To ensure comprehensive and diverse feature extraction while minimizing error propagation, we fuse the original single-step input fringes across M frequencies with the corresponding HDR outputs from the HDR Fringe Generation Module, and use them as input to this network. For accurate phase retrieval, the network is trained to predict the sine and cosine components of the fringe patterns. We use the results computed from the conventional T-exposure, M-frequency, N-step PSP method as ground truth, as defined in Eq. 6. The wrapped phase φ_{Net} and the absolute phase are recovered from the predicted components using Eq. 7 and Eq. 8, respectively. The absolute phase is subsequently mapped to 3D coordinates via a calibrated lookup table, as shown in Eq. 9. To supervise the training, we employ both MSE and MAE losses, as shown in Eq. 14.

$$\varphi_{Net} = \tan^{-1}\left(\frac{S_{pred}}{C_{pred}}\right) \tag{18}$$

The HDR Generation Module and the Phase Calculationis Module complement each other and are trained simultaneously, forming an end-to-end network that transforms low-quality LDR fringe patterns into high-quality phase information. Compared to a cascaded structure [31] of HDR generation followed by phase unwrapping, this integrated system is more stable and reduces cumulative error effects. Once trained, the network requires only first-step fringe patterns of M-frequency under the longest and shortest exposures during measurement to accurately recover the phase and complete 3D reconstruction.

In this paper, we leverage the fitting capability of deep networks to reduce the traditional T-exposure, M-frequency, N-step PSP method to only two exposures, M frequencies, and a single phase-shifting step. The proposed HDRSL-Net serves as a general measurement pipeline adaptable to various encoding strategies, such as the three-frequency heterodyne method.

IV. Experiments and Results Analysis

A. Experimental Setup and Dataset Preparation

1) System Setup: To validate the proposed method, we built a structured light measurement system consisting of a projector (DLP LightCrafter 6500; Texas Instruments, USA) and a camera (acA2040-120 μm; Basler, Germany) equipped with a 15 mm focal length lens. Both the projector and camera have a resolution of 1920×1080 pixels. The system is positioned 0.5–0.6 m from the measured object, covering a measurement area of approximately 0.2×0.18 m^2, as illustrated in Fig. 4. The highest fringe frequency is set to 64, corresponding to a 30-pixel period to match the projector resolution and ensure integer phase-shifting steps. System calibration is conducted using a four-frequency (1, 4, 16, 64) 12-step phase-shifting algorithm combined with Zhang's method [35]. A circular dot calibration board is placed at 20 positions within the measurement volume to compute pixel-wise absolute phase and 3D coordinates. Calibration parameters are obtained by fitting Eq. 9 using the least-squares method. HDRSL-Net is implemented in Python with PyTorch and trained on a workstation equipped with dual Intel Xeon CPUs (3.30 GHz), 64 GB RAM, and four NVIDIA RTX 3090 GPUs.

2) Dataset Preparation: To address the lack of large-scale HDR datasets for structured light in industrial scenarios, we constructed a dataset of 1700 groups of machined metal objects, as shown in Fig. 5. It includes various materials with high (e.g., aluminum, stainless steel, nickel alloy, brass) and

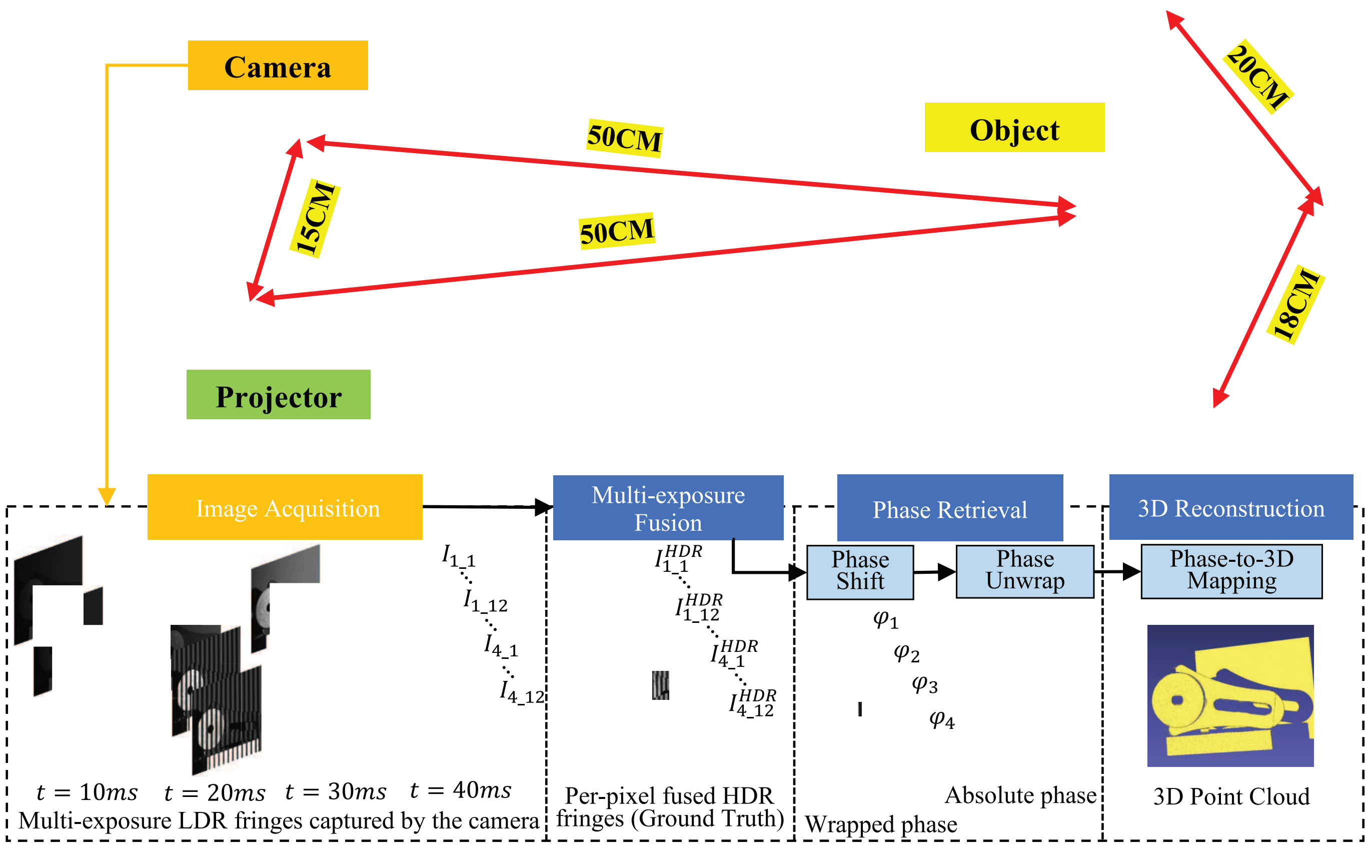

Fig. 4. Architecture of the HDR structured light system. The camera captures T M N fringe patterns, which are fused into HDR fringes (1 M N). Phase-shifting computes the sine and cosine components and the wrapped phase (1 M 1), which is then unwrapped to obtain the absolute phase for 3D reconstruction.

(a) (b) (c)

(d) (e) (f) (g)

Fig. 5. Representative shapes and materials from the 1,700-set metal dataset: (a)–(e) are high-reflectivity metals (aluminum alloy, stainless steel, nickel-plated alloy, aluminum, brass) requiring short exposures to avoid saturation, while (f) and (g) are low-reflectivity materials (magnesium oxide alloy, cast iron) requiring longer exposures to ensure adequate SNR.

low reflectivity (e.g., magnesium oxide alloy, cast iron), covering typical machining features such as flat/sloped surfaces, drill holes, chamfers, fillets, and standard parts like gears and bearing seats. Object sizes range from 2–12 cm in width and 1–8 cm in height. To facilitate robust phase unwrapping and effective feature learning, we employ four fringe frequencies $(1; 4; 16; 64)$ with 4 frequency scaling. Exposure time is selected based on reflectivity analysis: 10 ms avoids saturation on bright surfaces, while 40 ms provides sufficient SNR for dark surfaces. We adopt four exposure levels (10, 20, 30, and 40 ms) to fuse HDR fringe patterns. Each object sample contains 192 captured images (4 exposures 4 frequencies 12 phase steps). HDR fringes are synthesized using Eq. 11, and sine/cosine components $(S; C)$ are computed per frequency via Eq. 6. The first-step fringes at the shortest and longest exposures (2 4 1) are used as input. The corresponding HDR fringes (1 4 1) serve as GT for the HDR Generation Module, and the $(S; C)$ pairs (1 4 1) are used as supervision for the Phase Calculation Module.

To evaluate generalization on complex surfaces, we further test our method on a publicly available gypsum dataset [10] primarily containing human faces made of gypsum, which typically do not require HDR imaging. To simulate varying exposures and ensure compatibility, the original fringe intensities are scaled to one-third and twice their values to mimic under- and overexposure.

Both the metal and gypsum datasets were split into training, validation, and testing sets using an 8:1:1 ratio. The training parameters were set as follows: batch size of 6, an initial learning rate of 0.001 with cosine decay, and runs for 400 epochs.

B. Results and Analysis of HDR Reconstruction Pipeline

We evaluated our method on a metal dataset with high dynamic imaging requirements. When the surface reflectivity

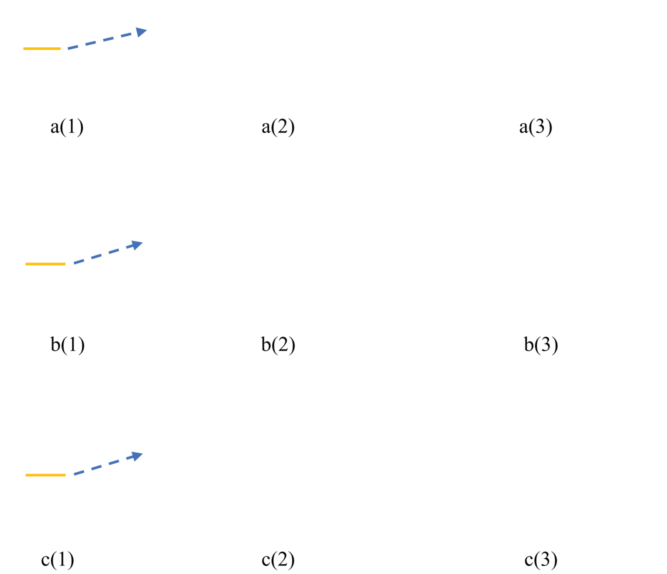

Fig. 6. Analysis of results in high reflectivity regions: (a) Imaging result with an exposure time of 10 *ms*, (b) Imaging result with an exposure time of 40 *ms*, (c) Output result of the HDR Generation Module, where (1) shows the fringe pattern, (2) the intensity curve, (3) comparison of the error curve against the 4-exposure HDR reference.

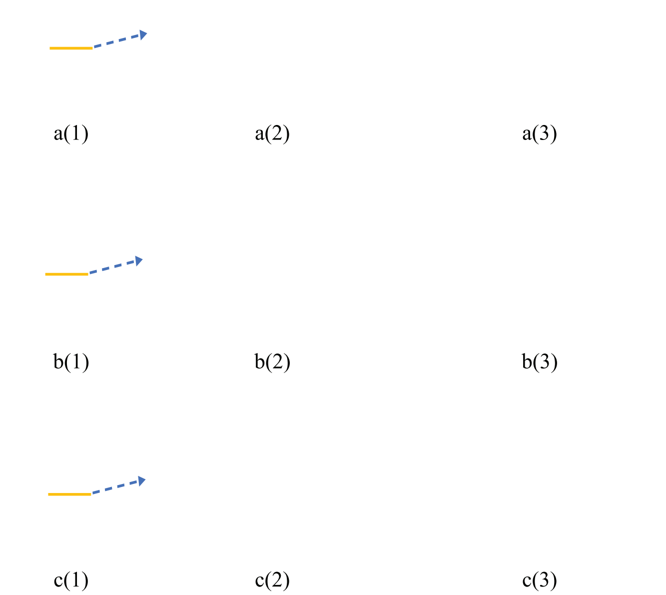

Fig. 7. Analysis of results in low reflectivity regions: (a) Imaging result with an exposure time of 40*ms*, (b) Imaging result with an exposure time of 10 *ms*, (c) Output result of the HDR Generation Module, where (1) shows the fringe pattern, (2) the intensity curve, (3) comparison of the error curve against the 4-exposure HDR reference.

of the measured object is high, the fringe pattern shows saturation, as shown in Fig. 6(b), with errors occurring in highly reflective areas and regions with strong projection intensity. Low exposure is needed for measurement, as illustrated in Fig. 6(a). The HDR generation module corrects the saturated fringe pattern into a HDR fringe pattern, achieving very low reconstruction error, as shown in Fig. 6(c). Our method effectively suppresses the saturated regions and accurately learns the feature distribution of the 4-exposure GT, resulting in a light intensity distribution that closely matches the true values, while retaining fringe patterns with phase information.

When the surface reflectivity is low, the captured fringe pattern has low light intensity, as shown in Fig. 7(b). This leads to a low SNR in the fringe pattern, increasing the error in phase unwrapping and causing higher HDR generation error. As shown in Fig. 7(a), higher exposure can effectively mitigate this issue. The HDR Generation Module in this paper effectively enhances the light intensity in the dark regions of the fringe pattern, as shown in Fig. 7(c), and achieves low HDR generation error.

The experiments demonstrate that the HDR Generation Module effectively learns the HDR fringe characteristics across 4-exposures. It can reconstruct the LDR fringe images captured under long (40 *ms*) and short (10 *ms*) exposures into HDR fringe images containing phase information, without needing to consider the specific properties of the measured object. This ensures accuracy while also improving the speed of HDR imaging.

We further evaluate HDRSL-Net's performance in phase calculation and 3D reconstruction. The predicted sine and cosine components (S, C) are used to compute the wrapped phase, which is then unwrapped to obtain the absolute phase for 3D surface reconstruction. The solving process for some metal objects is shown in Fig. 8. First, for metal parts with saturated fringe patterns, the measurement results are shown in Fig. 8(a). The HDR Generation Module accurately modulates the saturated regions (see column 2), and the Phase Calculation Module subsequently recovers the phase accurately (see column 3), further reconstructing the precise 3D point cloud

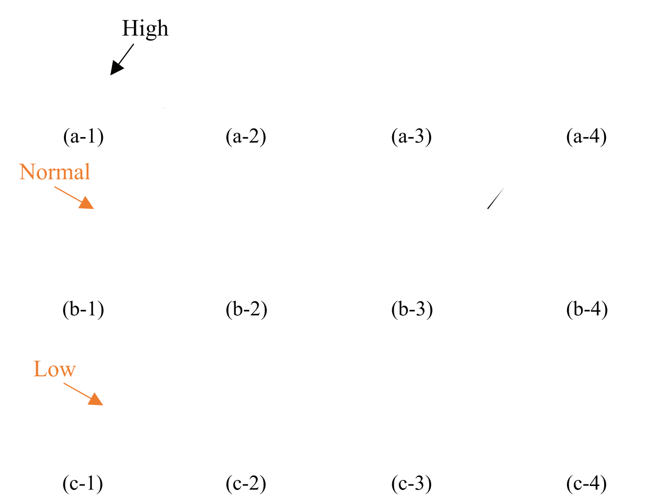

Fig. 8. Visualization of pipeline results on objects requiring HDR imaging. (a) High-reflectivity object; (b) Object with mixed reflectivity (high, low, normal); (c) Low-reflectivity object. For each case: (1) Objects with varying reflectivity; (2) is the HDR fringe generated by the HDR Generation Module; (3) is the absolute phase error map; (4) is the reconstructed 3D point cloud.

TABLE I

MAE COMPARISON OF SINE/COSINE COMPONENTS, WRAPPED PHASE, AND ABSOLUTE PHASE ON THE METAL DATASET (REPORTED AS MAE STD), ALONG WITH THE CORRESPONDING NUMBER OF PROJECTED IMAGES (T M N), MODEL SIZE (M PARAMETERS), AND TRAINING TIME (H)

Methods (exposure)	Projection Images	Model Size (M)	Training Time (h)	Sin (MAE $\pm$ std)	Cosine	Wrapped Phase	Absolute Phase
12-step PSP (10ms)	$1 \times 4 \times 12$	–	–	0.3395 ± 0.0787	0.3402 ± 0.0786	0.0567 ± 0.0871	0.1285 ± 0.3517
12-step PSP (40ms)	$1 \times 4 \times 12$	–	–	0.2764 ± 0.1256	0.2765 ± 0.1251	0.0571 ± 0.1063	0.1992 ± 0.7254
12-step PSP (10ms + 40ms)	$1 \times 4 \times 12$	–	–	0.6312 ± 0.0650	0.6312 ± 0.0652	0.0467 ± 0.0408	0.1159 ± 0.2745
6-step PSP (4-exposure)	$4 \times 4 \times 6$	–	–	0.5924 ± 0.0625	0.5923 ± 0.0631	0.0329 ± 0.0337	0.0690 ± 0.0264
6-step PSP (10ms)	$1 \times 4 \times 6$	–	–	0.8349 ± 0.0930	0.8339 ± 0.0926	0.1363 ± 0.1422	0.2426 ± 0.2865
4-step PSP (4-exposure)	$4 \times 4 \times 4$	–	–	0.7942 ± 0.0837	0.7918 ± 0.0839	0.0726 ± 0.0798	0.6617 ± 1.9715
4-step PSP (10ms)	$1 \times 4 \times 4$	–	–	0.9547 ± 0.0972	0.9525 ± 0.0972	0.1037 ± 0.0911	0.7192 ± 1.9867
FTP (10ms)	$4 \times 4 \times 1$	–	–	1.5873 ± 0.1592	0.7930 ± 0.1038	0.1893 ± 1.0370	1.1321 ± 0.0827
Y-FFC (40ms)	$1 \times 4 \times 12$	70	3	0.4300 ± 0.0854	0.4297 ± 0.0853	0.1803 ± 0.0972	0.1863 ± 0.1576
DC-UNet (40ms)	$1 \times 4 \times 12$	40	4.2	0.4472 ± 0.0932	0.4463 ± 0.0931	0.1762 ± 0.0976	0.1790 ± 0.1366
HDRNet (40ms)	$1 \times 4 \times 12$	35	3	0.4462 ± 0.1046	0.4460 ± 0.1047	0.1818 ± 0.1101	0.2315 ± 0.2649
Denoise Net (10ms)	$1 \times 4 \times 12$	31	17	0.0634 ± 0.0184	0.0553 ± 0.0171	0.0833 ± 0.0516	0.1753 ± 0.1982
HDRSL-Net (10ms + 40ms)	$2 \times 4 \times 1$	76	9	0.0449 ± 0.0175	0.0408 ± 0.0166	0.0630 ± 0.0480	0.1055 ± 0.1848

(see column 4). Fig. 8(c) shows the measurement of low-reflectivity objects, where HDR similarly enhances the SNR of fringe pattern and restores the phase and 3D point cloud through the Phase Calculation Module. Fig. 8(b) illustrates the measurement process of objects with normal, high, and low reflectivity. The generated HDR images accurately capture the fringe characteristics of objects with all three types of reflectivity, which are effectively reconstructed in 3D.

The experiments validated the effectiveness of the proposed HDR reconstruction pipeline for measuring objects with varying reflectivity. The HDR can efficiently modulate HDR fringe patterns containing phase information, while the Phase Calculation Module can extract the phase information from a single fringe pattern and further solve the 3D shape. Our pipeline only requires two low dynamic range fringe patterns under normal exposures to complete 3D reconstruction, improving the usability of structured light for 3D reconstruction, especially in industrial scenarios involving objects with high surface reflectivity variations.

C. Comparison of Existing Methods

The measurement accuracy increases with the number of steps. Using the results under 4 exposures, 4 frequencies and 12 steps as the baseline, we compared our method with traditional and deep learning methods on a metal dataset. The traditional methods include 12-step PSP method under 10 ms and 40 ms exposures, their fusion, as well as 6-, 4-, 3-, and 1-step FTP variants with 4-exposure. The deep learning methods consist of Y-FFC [29], HDRNet [28], and DC-UNet [30] for high reflectivity modulation, as well as a denoising network inspired by SP-CAN [31] for enhancing fringe patterns under low SNR conditions. The results are shown in Table. I. Firstly, our method achieves a Phase MAE of 0.1054, surpassing the 12-step PSP method under LDR fringes (0.1285 at 10 ms and 0.1992 at 40 ms) and approaching the performance of the 6-step PSP method with 4 exposures (0.069), while using only 8.3% of the projected images (reduced from 4 4 6 to 2 4 1). Secondly, from the analysis of Table. I, using multiple exposures with more projection steps (6 in this study) significantly improves phase accuracy. This is because, in the PSP method, each pixel has at least three effective samples, and multiple exposures are only effective with sufficient steps.

(a) (b)

(c) (d)

(e) (f)

(g) (h)

Fig. 9. Visualization of results from different algorithms on objects with varying reflectivity: (a) 12-step PSP with 4 exposures: absolute phase error, and 3D point cloud; (b) Fourier-based method with 4 exposures; (c) 12-step PSP at 10 ms; (d) Denoise Net; (e) 6-step PSP with 4 exposures; (f) Denoise Net; (g) 6-step PSP at 10 ms; (h) Proposed method.

As shown in Fig. 9, we evaluated the performance of different methods on objects with regions of normal, high, and low reflectivity. Our method shows minimal error in both bright and dark regions, approaching the performance of the 6-step method with 4 exposures. In contrast, the PSP method without multiple exposures exhibits higher errors in saturated and overly dark regions. Meanwhile, single-step Fourier transform-based methods are sensitive to object textures and often struggle to recover fine local surface details, as shown

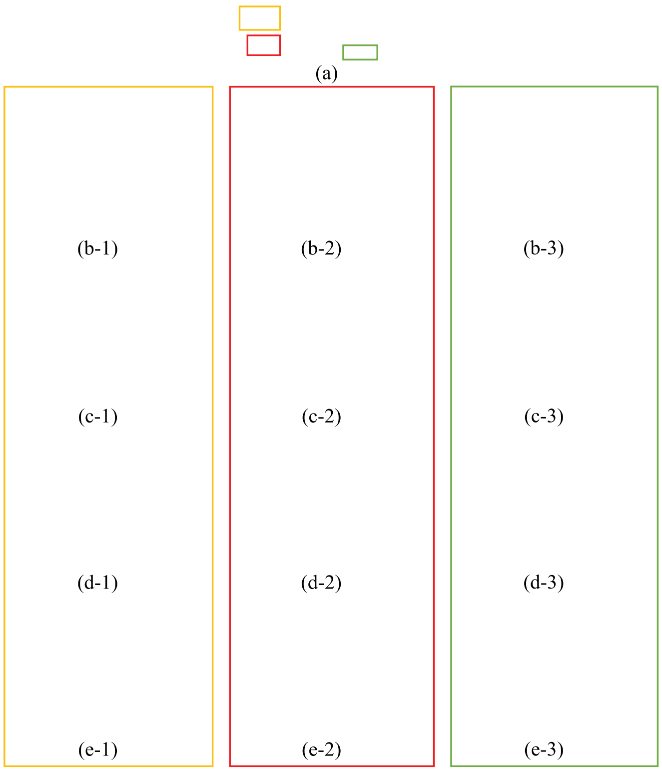

Fig. 10. 3D reconstruction results across regions of medium (1), low (2), and high (3) reflectivity: (a) Object with varying reflectivity, (b) 6-step PSP at 10 ms, (c) Denoise Net, (d) DC-UNet, (e) Proposed method.

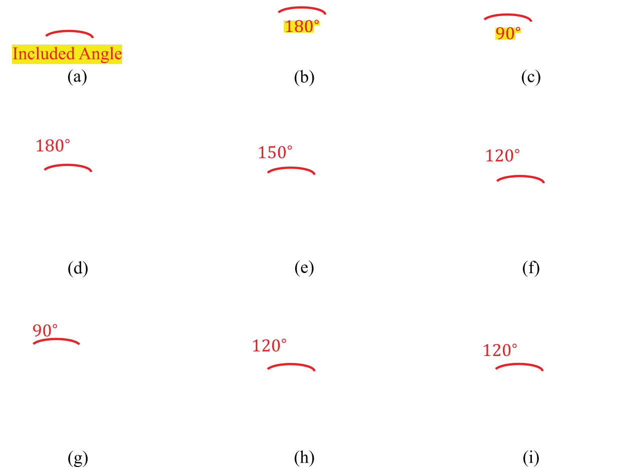

Fig. 11. (a) Object setup, (b–c) fringe patterns at 180 and 90 , (d–g) point clouds at 180 , 150 , 120 , and 90 , (h–i) results of four- and three-frequency PSP at 120 .

in Fig. 9(b). In contrast, our single-step method essentially fits temporal phase recovery via deep learning, making it robust to surface texture variations. Compared to direct HDR fusion with 10 ms and 40 ms exposures, our method generates richer HDR fringe features, achieves higher accuracy, and reduces phase steps to accelerate reconstruction. Deep learning-based light modulation methods struggle to balance bright and dark regions.

Fig. 10 presents the local point cloud reconstruction results for the three regions. Traditional PSP methods are significantly affected by noise in low-light regions, particularly at the boundaries between bright and dark areas. The denoising network, which uses low-illumination LDR images as input, performs well in high- and normal-reflectivity regions but exhibits significant errors in low-reflectivity areas due to insufficient structural information. The DC-UNet algorithm, which suppresses high-reflectivity input, performs well in low-reflectivity regions but struggles to reconstruct high-reflectivity areas. The results demonstrate that relying on LDR images from a single exposure is still insufficient for solving HDR measurement problems. Our method performs consistently across different regions, effectively reconstructing objects with varying surface reflectivity, and demonstrates the capability for HDR imaging and measurement.

Compared to deep learning algorithms such as DC-UNet, Y-FFC, and HDRNet, which are trained on simulated data and require less than 5 hours of training time, our method achieves smaller reconstruction errors in high-reflectivity regions and benefits from easier access to GT data, avoiding the generalization issues associated with simulated data. In comparison to the denoising-based method, our approach achieves higher accuracy and is more adaptable to objects with a wider range of surface reflectivity.

It should be noted that in our metal dataset, the shortest exposure (10 ms) effectively suppresses saturation, while the longest exposure (40 ms) improves the SNR. However, this exposure setting may not generalize to all scenarios. The proposed HDR framework is adaptable, with the minimum and maximum exposure times adjustable according to the specific measurement context.

D. Generalization and Robustness Evaluation

Due to the tendency of metals to produce strong interreflection and the ease with which flat surfaces can be fitted by deep networks, we further evaluate the generalization ability of HDRSL-Net on both highly reflective metal objects and a gypsum dataset with complex surface geometry.

1) Generalization on Interreflective Metals: We designed an angular structure composed of two aluminum alloy objects to evaluate the robustness of HDRSL-Net under interreflection conditions. As shown in Fig. 11(a), the lower region of the objects exhibits strong interreflection, while the upper region remains largely unaffected. The severity of interreflection increases as the angle between the objects decreases, as illustrated in Fig. 11(b) and (c). Measurements were conducted at four angles: 180 , 150 , 120 , and 90 . As shown in Fig. 11(d)–(g), HDRSL-Net successfully reconstructs the 3D shape at 180 and 150 . At 120 , reconstruction quality degrades with noticeable noise in the point cloud, and at 90 , severe interreflection leads to significant phase errors, preventing accurate reconstruction.

(a) (b) (c)

(d) (e) (f)

Fig. 12. (a) Low-exposure fringe, (b) Ground-truth HDR fringe, (c) Predicted phase by HDRSL-Net, (d) High-exposure fringe, (e) Predicted HDR fringe, (f) Phase error map.

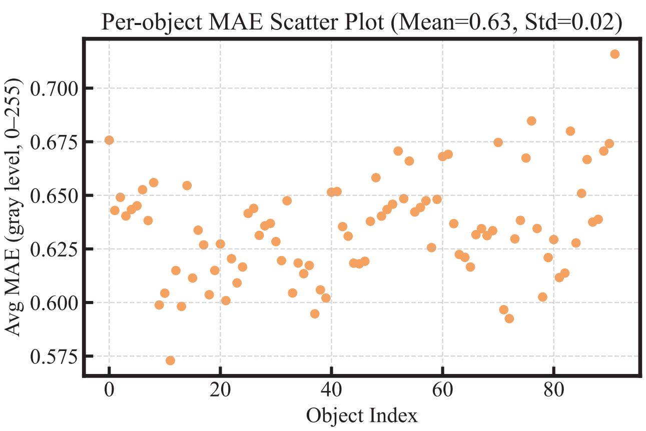

Fig. 13. Error distribution of HDR fringe reconstruction on the gypsum dataset.

Fig. 11(h) and (i) show the results of the four-frequency (1, 4, 16, 64) and three-frequency (4, 16, 64) 12-step PSP methods at the 120 configuration, respectively. Introducing the lowest frequency fringe significantly improves the system's robustness to interreflection and enhances the stability of initial phase unwrapping. HDRSL-Net learns from the four-frequency PSP strategy and exhibits strong robustness under intense global illumination. However, in cases of extremely strong interreflection or specular surfaces, HDRSL-Net may still encounter limitations. In such scenarios, more robust encoding strategies, such as multi-frequency heterodyne methods or more suitable frequency combinations, may be required. Notably, our framework is compatible with these strategies.

2) Generalization on the Gypsum Dataset: We conducted experiments on a gypsum face dataset. To simulate different exposure conditions, the fringe intensities were scaled to one-third and twice the original values, as shown in Fig. 12(a) and (d). The original fringe patterns were used as ground truth for the HDR generation module" as shown in Fig. 12(b). As shown in Fig. 12(e), HDRSL-Net effectively reconstructs HDR fringe patterns that closely resemble the originals, particularly around complex regions such as beards and eyes. Figure 13 presents the MAE across all objects, with a per-pixel error of 0.63 and a maximum error of only 0.72 under the 0–255 grayscale range. The high-quality reconstructed HDR fringes provide a reliable foundation for subsequent phase estimation.

TABLE II
PHASE ACCURACY ON THE GYPSUM DATASET (MAE STD, IN RADIANS)

Method	Wrapped Phase	Absolute Phase
6 step PSP	0.0157 ± 0.0020	0.0087 ± 0.0022
4 step PSP	0.0217 ± 0.0027	0.0123 ± 0.0029
3 step PSP	0.0898 ± 0.0119	0.0826 ± 0.0166
FTP	0.1151 ± 0.0319	0.0670 ± 0.0164
HDRSL-Net	0.0324 ± 0.0079	0.0223 ± 0.0128

We compare the phase accuracy of HDRSL-Net with traditional PSP methods (6-step, 4-step, 3-step) and single-step FTP on the gypsum dataset, using 12-step PSP as the baseline. As shown in Table II, HDRSL-Net achieves an absolute phase error of 0.0223, close to the 4-step PSP (0.0123), while using only 8 projected fringes compared to 16.

These results validate the generalization capability of HDRSL-Net to complex surfaces and broader data domains. Moreover, as the gypsum dataset differs in exposure settings from the metal dataset, the proposed pipeline demonstrates strong adaptability and broad applicability across diverse measurement scenarios.

E. Ablation Study

Our HDR Structured Light 3D Reconstruction Network consists of an HDR Fringe Generation Module, which incorporates feature distillation and attention modules, as well as Phase Calculation Module. Together, these components enable accurate HDR generation and 3D reconstruction capabilities. We conducted ablation experiments to verify the effectiveness of each component.

1) HDR Generation Module: As shown in Table. III, when the HDR Generation Module is not used and only phase analysis is performed on the LDR images, the MAE of the absolute phase increases to 0.1927, which is 183% higher than when the HDR module is included. While the Phase Calculation Module has been proven effective in analyzing fringe phases, it struggles to accurately resolve phases from LDR fringes that are affected by image saturation or low SNR. The HDR generation module enhances phase accuracy by converting LDR images into HDR images, capturing more phase information, and thus enabling more precise phase retrieval and 3D point cloud reconstruction.

2) Single Exposure LDR Fringe as Input: We compared the results of the pipeline when using only a single LDR input. As shown in Table. III, when using a low-exposure LDR image as input, the phase MAE decreased from 0.1729 to 0.1055, a reduction of 39.0%. When using a high-exposure LDR fringe as input, the phase MAE was 0.1978, a 46.7% reduction. As illustrated in Fig. 14(a) and (b), the HDR generation module struggles to reconstruct feature-rich HDR fringes from a single LDR image. The reconstruction errors for low-exposure input are concentrated in dark fringe regions, while for high-exposure input, errors are concentrated in bright regions. Since the experimental dataset contains a larger number of highly reflective objects, the low-exposure LDR

TABLE III

Ablation Study Analyzing the Impact of Input Fringe Pattern Types, Feature Distillation, Attention Guidance, and the HDR Generation Module on the MAE for Sine/coSine Components, Wrapped Phase, and Absolute Phase (MAE Std, in radians)

Methods	Sin	Cosine	Wrapped Phase	Absolute Phase
without HDR Generation Module	0.0584 ± 0.0196	0.0504 ± 0.0181	0.0825 ± 0.0547	0.1963 ± 0.2215
without Feature Distillation	0.0517 ± 0.0164	0.0445 ± 0.0148	0.0787 ± 0.0533	0.1563 ± 0.2004
without Attention Guided	0.0489 ± 0.0169	0.0422 ± 0.0150	0.0722 ± 0.0529	0.1390 ± 1.1833
Single exposure LDR Fringe Input (40ms)	0.0712 ± 0.0307	0.0655 ± 0.0369	0.0872 ± 0.0686	0.1978 ± 0.3209
Single exposure LDR Fringe Input (10ms)	0.0622 ± 0.0180	0.0555 ± 0.0167	0.0797 ± 0.0498	0.1728 ± 0.2143
HDRSL-Net	0.0449 ± 0.0174	0.0408 ± 0.0166	0.0630 ± 0.0480	0.1055 ± 0.1848

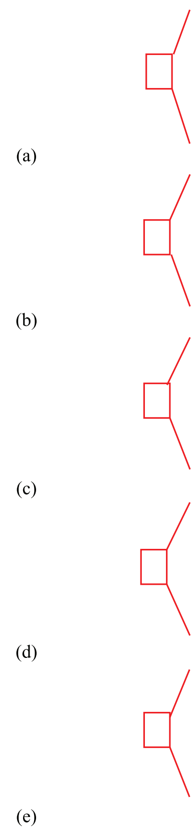

Fig. 14. Ablation results (from left to right): reconstructed fringe at $f = 1$, reconstruction error, 3D point cloud, and error region details. (a) Input: 10 ms exposure only; (b) Input: 40 ms exposure only; (c) Without attention module; (d) Without feature distillation; (e) Our full HDRSL-Net.

input results in lower phase error compared to high-exposure input. The experimental results suggest that inputting fringe patterns under both short and long exposure times effectively modulates both saturated and low-light fringe patterns, and the Phase Calculation Module can further resolve their phases.

Single-exposure input reduces accuracy but improves speed, making it suitable for time-sensitive applications. Moreover, reconstruction errors under low and high exposures are concentrated in dark and bright fringe regions, respectively. Adaptive exposure strategies may mitigate this issue and further improve reconstruction accuracy.

3) Feature Distillation and Attention Modules: As shown in Table. III, the feature distillation module reduces the phase error MAE from 0.1563 to 0.1055, representing an improvement of 32.5%. Fig. 14(d) demonstrates that the feature distillation module promotes the generation of higher-quality HDR images at different frequencies within the HDR generation module, reducing phase and 3D point cloud errors in high-reflectivity regions. Similarly, the attention module strengthens HDR generation features by training and fusing the weighted features of both bright and dark LDR images, improving the MAE from 0.1390 to 0.1055, a 24.1% improvement, as shown in Fig. 14(c). This leads to the generation of higher-quality HDR fringe patterns and more accurate phase retrieval and 3D point cloud results.

F. Quantitative Evaluation of 3D Reconstruction Accuracy

We conducted quantitative measurements on several standard objects, including two spheres with radii of 15.0086 0.002 mm and 12.6975 0.002 mm, a plate with a flatness of 0.015 mm, and a step object with a width of 1.5988 0.005 mm and a height of 2.0024 0.005 mm. Ground-truth were obtained via a coordinate measuring machine (CMM). As shown in Fig. 15, the spheres and plate are made of ceramic, while the step object is made of metal and requires HDR measurement due to its high surface reflectivity.

As illustrated in the third column of Fig. 15, we adopt different accuracy metrics for the three test objects. For the plate, we used the least squares method to fit an ideal plane to the point cloud, and flatness of the fitted surface was used as the error metric. For the spheres, the radius was estimated from the point cloud, and the absolute error was computed as jL_{pred} L_{GT}j, where L_{GT} is the CMM-measured data. For the step object, the width and height were obtained by projecting the point cloud along horizontal and vertical directions, and the absolute error was computed in the same manner. To ensure repeatability, each measurement was performed 10 times.

The measurement errors are presented in Table. IV. The measurement error of PSP methods generally decreases as the number of phase-shifting steps increases. For the ceramic spheres and plate, multiple exposures had minimal impact on the results. Our method achieved radius measurement errors of 41.3 m and 44.7 m, and a flatness of 41.7 m—approaching to the 4-step PSP method while using only 12.5% of the projection time. Since the surface is ceramic, multiple exposures did not significantly impact the results. Deep learning methods such as Y-FFC, HDRNet, DC-UNet and DenoiseNet showed limited accuracy in LDR scenarios.

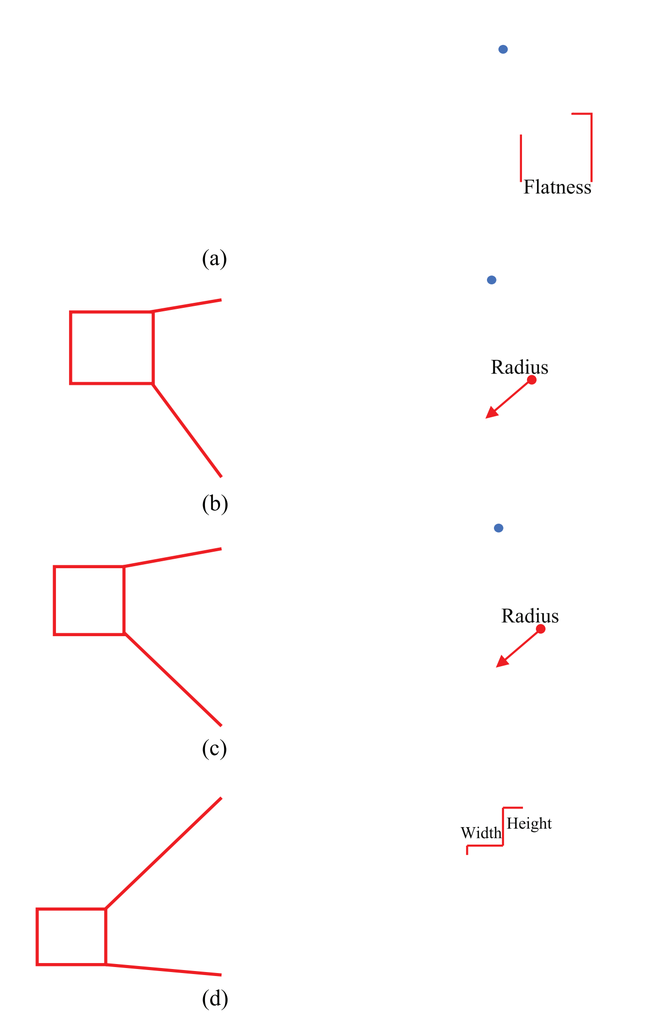

Fig. 15. From left to right are the fringe pattern, the reconstructed 3D point cloud, and the measurement metric. (a) A ceramic plate with a flatness of 0.015 mm; (b) A ceramic sphere with a diameter of 15.0086 0.002 mm; (c) A ceramic sphere with a diameter of 12.6975 0.002 mm; (d) A metallic staircase with a width of 1.5988 0.005 mm and a height of 2.0024 0.005 mm.

TABLE IV
MEASUREMENT ERROR OF EXISTING METHODS ON SPHERE, PLATE, AND METAL STEP OBJECTS (IN MM)

Methods	**Sphere1**	**Sphere2**	**Plate**	**Width**	**Height**
12-step PSP (4Exp)	0.0119	0.0198	0.0347	0.0217	0.0183
12-step PSP	0.0119	0.0211	0.0353	0.0369	0.0310
6-step PSP	0.0153	0.0238	0.0434	0.0393	0.0348
4-step PSP	0.0162	0.0248	0.0490	0.0495	0.0543
Y-FFC	0.0451	0.0424	0.0677	0.0709	0.0857
HDRNet	0.1023	0.1105	0.0692	0.0819	0.1050
DC-UNet	0.0774	0.0698	0.0593	0.0606	0.0634
Denoise Net	0.0778	0.0729	0.0460	0.0423	0.0549
HDRSL-Net	0.0413	0.0448	0.0417	0.0334	0.0306

For the metal step object, multiple exposures demonstrated clear advantages in handling highly reflective surfaces. Our method achieved width and height errors of 33.4 m and 30.6 m, respectively—matching the performance of the 6-step PSP method. Deep learning methods trained on synthetic data, such as Y-FFC, HDRNet, and DC-UNet, exhibited generalization issues. In contrast, the denoising-based DenoiseNet, leveraging low-exposure inputs, showed better performance on metal surfaces.

In terms of quantitative evaluation, our method successfully reconstructed the 3D shape and achieved sub-50 m measurement accuracy.

V. Conclusion

In this paper, we introduce deep learning into the HDR imaging-based structured light measurement system and propose a 3D reconstruction pipeline for objects with high dynamic reflectivity. The proposed approach comprises two end-to-end modules: an HDR Fringe Generation Module enhanced by feature distillation and attention mechanism, and a Phase Calculation Module. The former efficiently generates phase-preserving HDR fringes from short- and long-exposure LDR inputs, while the latter accurately retrieves phase using a single-step fringe.

For 3D reconstruction of objects with HDR surface reflectivity, our method addresses the low efficiency of traditional multi-exposure PSP approaches while maintaining comparable accuracy. On the metal dataset, our method achieved a phase error of 0.1050, outperforming other HDR deep learning-based methods and approaching the accuracy of the traditional 4-exposure 6-step PSP method (0.069), while taking only 8.3% of the projection time. Experimental results demonstrate the robustness of our approach under diverse object geometries, illumination levels, and challenging global lighting environments. In the quantitative evaluation, our method achieved sub-50 m accuracy on spheres, plate, and metal step objects, demonstrating its effectiveness in precise 3D reconstruction. Ablation experiments further demonstrated that the HDR Fringe Generation Module played a critical role in reconstruction, while the feature distillation and attention components effectively improved the reconstruction accuracy of this module.

Furthermore, we constructed an HDR imaging metal dataset comprising 1,700 samples of machined metal parts with diverse shapes, sizes, and materials, providing a valuable benchmark for HDR-based structured light 3D reconstruction. Overall, our method offers a fast and high-precision solution for structured light measurement under high dynamic reflectivity. As a data-driven method, it exhibits potential for achieving even greater accuracy with higher-precision ground truth.

Acknowledgment

The authors would like to thank all co-authors for their valuable contributions to this work.

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